

Education

Carnegie Mellon University , Pittsburgh, PA PhD in Computer Science. Thesis: Towards City-Scale Neural Rendering Committee: Deva Ramanan (chair), Jessica K. Hodgins, Martial Hebert, Jonathan T. Barron	08/24
Stanford University , Stanford, CA MS in Computer Science.	06/18
Stanford University , Stanford, CA BS in Computer Science , minor in Management Science and Engineering	06/12

Professional Experience

Research Scientist , NVIDIA, Seattle, WA - Exploring data-driven simulation and 3D deep learning topics in Sanja Fidler's group under Zan Gojcic	09/24 - Present
Research Scientist Intern , Meta, Pittsburgh, PA - Worked on real-time neural rendering in VR with Christian Richardt and Michael Zollhöfer	05/23-12/23
Research Intern , Argo AI, Pittsburgh, PA - Explored city-scale 3D reconstruction methods with Deva Ramanan and Francesco Ferroni	05/22-08/22
Engineering Manager and Technical Lead , Palantir Technologies, New York, NY - Led R&D efforts to improve Palantir's Foundry data platform: - Created interactive search and analysis infrastructure that evolved into one of the primary user interfaces of Palantir's Foundry data lake platform deployed across the company's customer fleet - Invented full-text search system that powers Palantir's internal log analysis infrastructure and a significant part of the Foundry data lake platform - Led inception and rollout of one of Palantir's flagship data analytics products - Managed the technical implementation of a variety of high-profile customer engagements in the commercial sector (leading teams of 10-100+ individuals) Supervised and mentored a team of 7+ direct reports	08/12-08/19
Engineering Intern , Addepar, Mountain View, CA - Worked on developing a software platform to help private financial advisors and family offices perform quantitative analysis and interface with their clients	04/11-09/11
Research Assistant , Stanford University, Stanford, CA - Conducted semantic segmentation research under Professor Daphne Koller and Dr. M. Pawan Kumar	06/10-12/10

Teaching Experience

- Graduate Student Instructor, Dept. of Computer Science, Carnegie Mellon University** 09/20-06-21
- Teaching assistant for 15-640 (Distributed Systems) and inaugural 17-700 class (Data Science and Machine Learning at Scale)
- Project Mentor, Dept. of Computer Science, Carnegie Mellon University** 09/20-12/21
- Designed a term project for 15-821 (Mobile and Pervasive Computing)
 - Met and provided guidance to project members on a regular basis

Publications (Reverse Chronological Order)

- [1] Zhangjie Wu*, Yuxuan Zhang*, **Haithem Turki**, Xuanchi Ren, Jun Gao, Mike Zheng Shou, Sanja Fidler, Zan Gojcic*, Huan Ling*, [“DIFIX3D+: Improving 3D Reconstructions with Single-Step Diffusion Models.”](#) To appear in CVPR 2025.
- [2] **Haithem Turki**, Vasu Agrawal, Samuel Rota Bulò, Lorenzo Porzi, Peter Kotschieder, Deva Ramanan, Michael Zollhöfer, Christian Richardt, [“HybridNeRF: Efficient Neural Rendering via Adaptive Volumetric Surfaces.”](#) In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024 (highlight).
- [3] Li Ma, Vasu Agrawal, **Haithem Turki**, Changil Kim, Chen Gao, Pedro Sander, Michael Zollhöfer, Christian Richardt, [“SpecNeRF: Gaussian Directional Encoding for Specular Reflections.”](#) In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024 (highlight).
- [4] **Haithem Turki**, Michael Zollhoefer, Christian Richardt, Deva Ramanan, [“PyNeRF: Pyramidal Neural Radiance Fields.”](#) In Proceedings of the 37th Conference on Neural Information Processing Systems (NeurIPS), 2023.
- [5] Shilpa George, **Haithem Turki**, Ziqiang Feng, Thomas Eiszler, Deva Ramanan, Padmanabhan Pillai, Mahadev Satyanarayanan, [“Low-Bandwidth Self-Improving Transmission of Rare Training Data.”](#) In Proceedings of the 29th Annual International Conference on Mobile Computing And Networking (MobiCom), 2023.
- [6] Shilpa George, **Haithem Turki**, Ziqiang Feng, Thomas Eiszler, Deva Ramanan, Padmanabhan Pillai, Mahadev Satyanarayanan, [“Edge-based Privacy-Sensitive Live Learning for Discovery of Training Data.”](#) In Proceedings of the 1st International Workshop on Networked AI Systems (NetAISys '23).
- [7] **Haithem Turki**, Jason Y. Zhang, Francesco Ferroni, Deva Ramanan. [“SUDS: Scalable Urban Dynamic Scenes.”](#) In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023.
- [8] Mahadev Satyanarayanan, Ziqiang Feng, Shilpa George, Jan Harkes, Roger Iyengar, **Haithem Turki**, Padmanabhan Pillai, [“Accelerating Silent Witness Storage,”](#) in IEEE Micro, Nov.-Dec. 2022.
- [9] **Haithem Turki**, Deva Ramanan, Mahadev Satyanarayanan, [“Mega-NeRF: Scalable Construction of Large-Scale NeRFs for Virtual Fly-Throughs.”](#) In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022.
- [10] Shilpa George, Thomas Eiszler, Roger Iyengar, **Haithem Turki**, Ziqiang Feng, Junjue Wang, Padmanabhan Pillai, Mahadev Satyanarayanan, [“OpenRTIST: End-to-End Benchmarking for Edge Computing,”](#) in IEEE Pervasive Computing, vol. 19, no. 4, pp. 10-18, 1 Oct.-Dec. 2020
- [11] Mahadev Satyanarayanan, Thomas Eiszler, Jan Harkes, **Haithem Turki** and Ziqiang Feng, [“Edge Computing for Legacy Applications,”](#) in IEEE Pervasive Computing, 1 Oct.-Dec. 2020
- [12] M. Pawan Kumar, **Haithem Turki**, Dan Preston and Daphne Koller, [“Parameter Estimation and Energy Minimization for Region-Based Semantic Segmentation,”](#) in IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 37, no. 7, pp. 1373-1386, 1 July 2015.

- [13] M. Pawan Kumar, **Haithem Turki**, Dan Preston and Daphne Koller, "[Learning specific-class segmentation from diverse data](#)," 2011 International Conference on Computer Vision.

Patents

- [1] **Haithem Turki**, Robert Fink, Amr Al Mallah. "[Systems and Methods for Indexing and Searching](#)." US Patent 2019/0347343 A1, filed June 8, 2018, and issued November 14, 2019.
- [2] **Haithem Turki**, Sander Kromwijk, Stephen Cohen, Yixun Xu, Feridun Arda Kara. "[Systems and Methods for Constraint Driven Database Searching](#)." US Patent 2018/293239 A1, filed April 11, 2017, and issued October 11, 2018.

Technical Reports

- [1] Ziqiang Feng, Shilpa George, **Haithem Turki**, Roger Iyengar, Padmanabhan Pillai, Jan Harkes, Mahadev Satyanarayanan. "[Improving Edge Elasticity via Decode Offload](#)." CMU-CS-21-139, September 2021.

Invited Talks

Towards City-Scale Neural Rendering

01/24-02/24

- Stanford Vision Lab
- Waymo Research (virtual)
- Google (virtual)
- NVIDIA Toronto AI Lab (virtual)

Mega-NeRF: Scalable Construction of Large-Scale NeRFs for Virtual Fly-Throughs

04/22

- Argo AI

Additional Information

- **Reviewer:** CVPR, ACCV, ECCV, ICCV, SIGGRAPH, SIGGRAPH Asia, AAAI, Transactions on Image Processing
- **Volunteer Activities:** Alchemist Accelerator (mentor and investor)
- **Languages:** Fluent in French and Arabic. Currently learning Japanese and German.
- **Other Qualifications:** Top Secret security clearance (expired). FINRA Series 65. Graduate of the Stanford StartX startup accelerator.